Exercise 13 :

1. Write the discretized form of the equations (42) for the zonal and meridional components using the grid shown in the previous slide.
2. Derive the non-dimensional Courant number that depends on the vertical viscosity
3. Using the vertical eddy viscosity shown fr:eddy-viscosity-diffusivity, what would be your choice of Δz and Δt to ensure stability?

	Molecular, at salinity = 35	Eddy: horizontal (along-isopycnal)	Eddy: vertical (diapycnal)
Viscosity	1.83×10^{-6} m^2/sec at 0 °C 1.05×10^{-6} m^2/sec at 20 °C	10^2 to 10^4 m^2/sec	10^{-4} m^2/sec
Thermal diffusivity	1.37×10^{-7} m^2/sec at 0 °C 1.46×10^{-7} m^2/sec at 20 °C	10^2 to 10^4 m^2/sec	10^{-5} m^2/sec
Haline diffusivity	1.3×10^{-9} m^2/sec	10^2 to 10^4 m^2/sec	10^{-5} m^2/sec

1. $$\frac{\partial U_E}{\partial t} - fV_E = A_v \frac{\partial^2 U_E}{\partial z^2}$$

$$\frac{\partial V_E}{\partial t} + fU_E = A_v \frac{\partial^2 V_E}{\partial z^2}$$

use the grids : $z_k = z_0 - k\Delta z$ and $t_i = t_0 + n\Delta t$

apply forward Euler in time and centred differences in space

time derivative : $\frac{\partial U}{\partial t} \approx \frac{U_k^{n+1} - U_k^n}{\Delta t}$

vertical diffusion : $\frac{\partial^2 U}{\partial z^2} \approx \frac{U_{k-1}^n - 2U_k^n + U_{k+1}^n}{\Delta z^2}$

zonal : $\frac{\partial U_E}{\partial t} - fV_E = A_v \frac{\partial^2 U_E}{\partial z^2}$

$$\frac{U_k^{n+1} - U_k^n}{\Delta t} - fV_{E_k}^n = A_v \frac{U_{k-1}^n - 2U_k^n + U_{k+1}^n}{\Delta z^2}$$

$$U_k^{n+1} = U_k^n + \Delta t \left[fV_{E_k}^n + A_v \frac{U_{k-1}^n - 2U_k^n + U_{k+1}^n}{\Delta z^2} \right]$$

meridional : $\frac{\partial V_E}{\partial t} + fU_E = A_v \frac{\partial^2 V_E}{\partial z^2}$

$$\frac{V_k^{n+1} - V_k^n}{\Delta t} + fU_{E_k}^n = A_v \frac{V_{k-1}^n - 2V_k^n + V_{k+1}^n}{\Delta z^2}$$

$$V_k^{n+1} = V_k^n + \Delta t \left[-fU_k^n + A_v \frac{V_{k-1}^n - 2V_k^n + V_{k+1}^n}{\Delta z^2} \right]$$

2. Stability is governed by the Courant Number : $C = \frac{A_v \Delta t}{\Delta z^2}$ ↳ diffusion term

for the forward Euler scheme : $C \leq 1/2$

3. $A_v = 10^{-4}\ m^2/s$ (vertical eddy viscosity)

stability: $C \leq 1/2$

$$C = \frac{A_v \Delta t}{\Delta z^2}$$

$$\Delta t \leq \frac{\Delta z^2}{2A_v} = \frac{\Delta z^2}{2 \times 10^{-4}}$$

check the given values: $\Delta z = 2m$, $\Delta t = 160s$

$$C = \frac{10^{-4} \times 160}{2^2} = \frac{0.016}{4} = 0.004 \leq \frac{1}{2}$$

this is highly stable so the grid is reasonable

maximum Δt for $\Delta z = 2m$

$$\Delta t_{max} = \frac{\Delta z^2}{2A_v} = \frac{4}{2 \times 10^{-4}} = 20000\,s = 5.5\ hrs$$

minimum Δz for $\Delta t = 160s$

$$\Delta z_{min} = \sqrt{2 A_v \Delta t} = \sqrt{2(10^{-4})(160)} = 0.18\ m$$

$\Delta t = 160s$ is very conservative and the scheme will still be stable for larger Δts

with $A_v = 10^{-4}\ m^2/s$, $\Delta t = 160s$ and $\Delta z = 0.18\ m$, the scheme is very much stable